

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded	
	Summer-Term Examination 2024-25	
	S.Y. B. Tech.	Chemical Engineering
	Chemical Engineering Thermodynamics-II	PCC-CH202/205
Date: 24.06.2025	Time: 14.30-17.30	Max. Marks : 100

- Note:
- Attempt all questions
 - Assume suitable data if necessary
 - Use of non-programmable calculator is permitted.

Q.N.	Question	Marks
Q.1	Sketch and explain in detail P-x-y diagram for binary mixture system. Also indicate equilibrium distribution diagram.	10
Q.2	What do you mean by partial molar properties? List out and explain the methods for determination of these properties in the solution thermodynamics.	10
Q.3	Define Excess properties in the thermodynamics and derive suitable expressions for excess Gibb's free energy.	10
Q.4	The standard heat of formation and standard free energy of formation of ammonia at 298 K are -46100 J/mol and -16500 J/mol respectively. Calculate the equilibrium constant for reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ at 500 K. Assume the std. heat of reaction is constant in between 298 to 500 K.	10
Q.5	Deduce phase equilibrium criteria for single component system. State criteria of stability.	10
Q.6	Calculate the equilibrium constant at 298 K of the reaction $\text{N}_2\text{O}_4(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$ Given that the standard free energies of formation at 298 K are 97,540 J/mol for N_2O_4 and 51,310 J/mol for NO_2 .	10
Q.7	Deduce Gibb's Duhem equation and state their applications.	10
Q.8	Explain in detail Raoult's law, Lewis Randall Rule and Henry law in the solution thermodynamics	10
Q.9	Define activity coefficient and explain in detail any activity coefficient model for the estimation of activity coefficient in binary solution system.	10
Q.10	Derive Gibb's Duhem equation. State its applications in solution thermodynamics.	10



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
Summer Term Examination
Academic Year: 2024-25

Class: Second Year

Course: Heat Transfer

Course Code: PCC-CH204

Date: 20.06.2025

Time: 14:30 to 17:30

Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- Solve any ten questions.

Q.N.			Marks	CO	BT
Q.1	a	Derive an equation for rate of heat flow through a cylinder.	10	CO2	BT6
Q.2	a	Define the following terms (a) Efficiency of Fins (b) Black body (c) Hydraulic diameter (d) Thermal conductivity (e) Critical insulation thickness (f) Overall heat transfer coefficients (g) Economy of evaporator (h) Emissivity (i) Intensity of radiation (j) Fouling factor	10	CO1,4,5	BT1,2
Q.3	a	A Furnace wall made up of steel plate 10 mm thick lined on inside with silica brick 150 mm thick on the outside with magnesite brick 150 mm thick. The temperature on the inside edge of wall is 973 K and on the outside is 288 K. Calculate the quantity of heat passed in W/m^2 . It is required to reduce the heat flow to $1163 W/m^2$ by means of an air gap between steel plate and magnesite brick. Estimate the width of this air gap. Thermal conductivities in W/mK are 16.86, 1.75, 5.23 and 0.33 for steel, silica brick, magnesite brick and air respectively.	10	CO2	BT3,5
Q.4	a	Calculate the heat transfer coefficient for fluid flowing through a tube having inside diameter of 40 mm at a rate of 5500 kg/hr. Assume that the fluid is being heated. Properties of fluid at mean bulk temperature: Viscosity = $0.004 Ns/m^2$, Density = $1.070 gm/cm^3$, Specific heat = $2.72 kJ/kg K$, Thermal conductivity = $0.256 W/mK$	10	CO3	BT3

Q.5	a	Compare the co-current and counterflow arrangements in heat exchangers	5	CO4	
	b	Explain the different laws of radiation along with their equations.	5	CO5	BT2,
Q.6	a	Derive an expression for the log mean temperature difference (LMTD). Explain the significance of it.	10	CO2	BT3
Q.7	a	A counter flow double pipe heat exchanger operates with hot water flowing inside the inner pipe and a polymer fluid flowing in the annular space between the two pipes. The water flowrate is 2 kg/s and it enters at a temperature of 90 °C. The polymer enters at a temperature of 10 °C and leaves at a temperature of 50 °C while the water leaves the exchanger at a temperature of 60 °C. Calculate the overall heat transfer coefficient by LMTD and ϵ -NTU method expressed in $W/m^2\text{ }^\circ C$, if the area for the heat exchanger is 20 m^2 . Data: $NTU=0.9$, Specific heat of water = $4.18\text{ kJ/kg }^\circ C$	10	CO3	BT5
Q.8	a	What is ϵ -NTU method? Derive an equation that illustrates the relationship between these parameters.	10	CO1	BT6
Q.9	a	Define the modes of heat transfer along with their respective laws.	5	CO1,2	BT2,3,6
	b	Write down the design procedure of finding out the heat transfer by convection.	5	CO4	BT6
Q.10	a	Water at the rate of 3.8 kg/s is heated from 38 to 55 °C in a shell and tube heat exchanger. On the shell side one pass is used with water as the heating fluid, 1.9 kg/s entering the exchanger at 93 °C. The overall heat transfer coefficient is $1419\text{ W/m}^2\text{ }^\circ C$, and the average water velocity in the 1.9 cm diameter tubes is 0.366 m/s. Because of space limitations the tube length must not be longer than 2.5 m. Calculate the number of tube passes, the number of tubes per pass, and the length of the tubes, consistent with this restriction. Data: $C_p=4.18\text{ kJ/kg }^\circ C$ Correction factor $F=0.88$	10	CO3,4	BT3,5
Q.11	a	An evaporator is operating at atmospheric pressure. It is desired to concentrate a feed from 4 % solute to 21 % solute (by weight) at a rate of 6000 kg/h. Dry saturated steam at a pressure corresponding to the saturation temperature of 399 K is used. The feed is at 298 K and the boiling point rise (boiling point elevation) is 5 K. The overall heat transfer coefficient is $2350\text{ W/m}^2\text{ K}$. Calculate the economy of evaporator and the area of heat transfer to be provided.	10	CO3,4	BT3,4

		Data: Latent heat of condensation steam at 399 K is 2185 kJ/kg Latent heat of vaporisation at 373 K is 2257 kJ/kg Specific heat of feed is 4.187 kJ/kg K			
Q.12	a	Calculate the rate of heat loss from a 10 m long horizontal steam pipe, 50 mm OD when carrying steam at 800 kN/m ² . The temperature of the surrounding atmosphere is 290 K. Data: Emissivity, $\epsilon = 0.85$ and $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$ The convective heat transfer coefficient is given by $h_c = 1.64 (\Delta T)^{0.25} \text{ (W/m}^2\text{K)}$ Steam is saturated at 800 kN/m ² and 443 K (170 °C).	10	CO1,4	BT3,5

Data:

Dittus Boelter Equation:

$Nu = 0.023 * Re^{0.8} * Pr^n$, where $n = 0.4$ for heating and 0.3 for cooling.

Sieder-Tate Equation:

$Nu = 0.027 * Re^{0.8} * Pr^{(1/3)} * (\mu_b / \mu_w)^{0.14}$

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand and analyze various modes of heat transfer
CO 2	Understand and calculate temperature profiles in a slab, cylinder, and sphere at steady state in a conduction
CO 3	Understand and calculate heat transfer coefficients in various equipments like double pipe heat exchangers, shell and tube heat exchangers, plate heat exchangers, condensation, evaporation, agitated tanks
CO 4	Understand and calculate heat duty, outlet temperatures, pressure drops, area required for various equipment like double pipe heat exchangers, shell and tube heat exchangers, plate heat exchangers
CO 5	Understand and analyze basics of radiation

SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

S.Y. B. Tech.
Chemical Engineering
Thermodynamics-I

Summer Term Examination 2024-25

Chemical Engineering
PCC CH203

Date: 23/06/2025

Time: 14.30 to 16.30

Max. Marks : 100

- Note:
- Attempt any five questions
 - Assume suitable data if necessary.
 - Use of non-programmable calculator is permitted.
 - This paper contains 01 page.

Q.N.	Question	Marks													
Q-1	a. Define the term activity and standard state in thermodynamics. Write the expression giving an effect of temperature and pressure on the activity.	10													
	b. What do you mean by equation of state? Using the Virial equation calculate the molar volume and compressibility factor of isopropanol vapour at 473 K and 10 bar . The Virial coefficients are $B = -3.88 \times 10^{-4} \text{ m}^3/\text{mol}$ and $C = -2.6 \times 10^{-8} \text{ m}^6/\text{mol}^2$.	10													
Q-2	Explain in detail following terms from thermodynamics 1. Type of thermodynamics systems: Open, Closed and isolated system 2. Reversible and irreversible process 3. State and path function 4. Concept of enthalpy	5x4 =20													
Q-3	a. Estimate fugacity for pure gas component using residual volume, compressibility factor and also using equation of state.	10													
	b. Derive first law of thermodynamics for flow process with proper example	10													
Q-4	a. Define Helmholtz free energy, degree of freedom and Gibb's phase rule.	10													
	b. Derive Gibb- Helmholtz's equation law of thermodynamics for flow process. List the limitations of first law of thermodynamics	10													
Q-5	a. Write Maxwell equations with its significance and derive the Clapeyron equation for pure fluids. Also list its application in thermodynamics.	10													
	b. Define residual thermodynamic properties and obtain an expression for residual entropy and enthalpy for pure fluid.	10													
Q-6	a. Give various statements of second law of thermodynamics. Explain the Carnot heat engine cycle in details.	10													
	b. Estimate the fugacity of gaseous propane at 12 bar and 310 K using the following data. <table><tr><td>P, bar</td><td>1.7</td><td>3.4</td><td>6.8</td><td>10.2</td><td>11.7</td><td>13.6</td></tr><tr><td>V, m³/kg</td><td>0.33</td><td>0.16</td><td>0.075</td><td>0.045</td><td>0.038</td><td>0.021</td></tr></table> Given that vapour pressure of propane at 310 K is 13 bar.	P, bar	1.7	3.4	6.8	10.2	11.7	13.6	V, m ³ /kg	0.33	0.16	0.075	0.045	0.038	0.021
P, bar	1.7	3.4	6.8	10.2	11.7	13.6									
V, m ³ /kg	0.33	0.16	0.075	0.045	0.038	0.021									
Q-7	a. Discuss in detail the vapour compression cycle with its significance. How is selection of refrigerant made?	10													
	b. Give brief description about the Hess's law of heat summation and determine the temperature change of the oil in which it flows at a rate of 1000 kg/min from open reservoir at the top of hill 400 m in height to another reservoir at the bottom of hill. Heat is supplied to the oil on its way at the rate of 1800 kJ/min and work is supplied by a 1 hp pump. The mean specific heat capacity of oil is 3.35 kJ/ Kg K.	10													